

ANDREY GOLOVANOV

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Education

University of Waterloo, Waterloo, ON

Bachelor of Applied Science in Computer Engineering

April 2026

Waterloo, Ontario

Technical Skills

Languages: C/C++, C#, Java, Python, VHDL, Verilog

Developer Tools: Visual Studio, VSCode, Android Studio, GoogleTest, Shell/Bash, VMware, Vivado

Technologies: Linux, STM32, Jenkins, GitHub, Azure, Docker, OpenGL, QT

Experience

Virtek Vision International

September 2025 – December 2025

Software Developer Intern

Waterloo, Ontario

- Engineered and optimized a **C# localization framework** for laser projection software, enabling accurate multilingual UI rendering and regional configuration.
- Built a reusable **DataGrid** and a separate **module-level injection mechanism**, allowing features such as sorting and deletion to be added without core code changes.
- Shaped architectural decisions around **model loading**, improving runtime module management.

RMUS Canada (now Remote Robotic)

January 2025 – August 2025

Software Developer Intern

Mississauga, Ontario

- Achieved **300% improvement** in real-time point cloud rendering by migrating to **OpenGL** and optimizing shaders, multithreading, and frustum culling.
- Cut **Leica LiDAR** bandwidth by **70%** via **delta encoding**, **LZ4 compression**, and **clustering**.
- Built a real-time **dynamic/static octree** in **PCL** managing **1M+ points** on resource-constrained systems.
- Developed and deployed **cross-platform control and visualization tools** for **Boston Dynamics Spot** using **CMake** and **Docker**, targeting **Windows**, **Linux**, and **Android**.

Dell Technologies

January 2024 – April 2024

Software Developer Intern

Ottawa, Ontario

- Built critical logging infrastructure for **5G ORAN** in a **large-scale C++ codebase**, improving observability by **25%** and reducing debugging time by **15%**.
- Adapted a **Python ASN.1** simulator for **3GPP** messages, verifying accuracy and reliability.
- Delivered a **live demo** to 80+ engineers, showcasing integration capabilities and practical use cases.
- Led **Agile ceremonies** including daily stand-ups, sprint planning, retrospectives, and backlog refinement.

Ford Motor Company

September 2022 – December 2022

Software Developer Intern

Kanata, Ontario

- Achieved **90% test coverage** in **C++** using **GTest** and mocking techniques.
- Independently mastered **OpenSSL**, **GNU Make**, and **CI/CD** within two weeks, accelerating delivery.
- Configured **Jenkins** for automated testing and wrote **Bash scripts** on **Linux**, saving **2 hours/week**.

Projects

Real-Time Operating Systems Kernel | C, STM32IDE

August 2024

- Developed a **real-time OS kernel** for STM32 Nucleo F401RE, supporting preemptive multitasking.
- Integrated **dynamic memory management** via the buddy system, enabling flexible task allocation.
- Designed and implemented an **Earliest Deadline First (EDF) scheduler** to meet real-time requirements.

Hot-Swappable Yield-Aware Dynamic Reconfiguration Architecture | C, FPGA May 2025 – Present

- Awarded Technical Excellence Award (2026)** for engineering a custom Linux Kernel Module enabling hot-swappable bitstreams with **80ms reconfiguration latency**.
- Developed the **core kernel** for the HYDRA FPGA runtime, allowing hardware logic to be managed as dynamic software tasks via CLI commands.
- Architected **Python integration** support, enabling high-level dynamic hardware accelerator management.